

Task 1:

A software company is developing an Online Food Delivery Payment System where customers can pay through EasyPaisa or JazzCash. Create an abstract class named PaymentSystem containing two pure virtual functions, pay() and refund(). Then create two derived classes, EasyPaisa and JazzCash, that implement these functions in different ways. In the EasyPaisa class, apply a 2% service fee during payment calculation, while in the JazzCash class, apply a 5% service fee. Display complete transaction details including the original amount, service fee or charges, and the final amount after deducting the service fee. In the refund() function, display the refund amount after deducting the service fee. In the main() function, use a base class pointer to call both payment methods, and demonstrate runtime polymorphism by switching the pointer between both derived classes. Print proper headings for each customer transaction and use meaningful values in the output.

Note: You should take amount from user in class using method. And necessary variables will be declared inside base class.

```
=====
=====
=====
=====
```

Task 2:

A university is creating a Student Fee Collection System where students can submit fee through BankTransfer or CreditCard methods. Create an abstract class FeeSystem with pure virtual functions submitFee() and returnFee(). Then create derived classes BankTransfer and CreditCard that override both functions. In BankTransfer, deduct a fixed transaction fee of Rs.100 from the submitted amount. In CreditCard, add a 3% processing charge on fee payment. Display full fee receipt including submitted fee, extra charges, total paid, and refund details. In main function, use base class pointer to access both classes. Show runtime polymorphism by changing object addresses.